Session 2: Dashboarding

Before you come to class

This pre-read takes about 25 minutes. Its big idea is simple: finding a number is one skill, and getting other people to act on it is another. A dashboard is how you do the second one.

Read it once, slowly. You don't need Google Sheets open. Notice the questions to ask *before* you build anything, and the traps to avoid when you read numbers. In class, we will build a real dashboard from Rapido ride data.

Two words you will see often. A **dashboard** is one screen that shows the few numbers one person needs to make one decision. A **KPI** (Key Performance Indicator) is a metric the team has agreed matters most, like "rides completed per day".

Our examples use **Rapido**, a bike-taxi app. A *rider* is the customer who books a ride. A *captain* is the driver who takes it.

1. Finding the answer is one skill, sharing it is another

Many people can find a number. Far fewer can make a busy person understand it and act on it. Data work is really three separate skills:

1. **Analyse and decide.** Dig into the data and work out what is going on.
2. **Choose what to track.** Pick the few metrics that matter for *your* role. (This is what Pre-read 1 was about.)
3. **Convey it to others.** Show it so a **stakeholder** (anyone who uses your analysis or is affected by it) can understand it in seconds and act.

A dashboard needs all three. Skill 1 alone gives you a pile of charts that only you understand.

The book *Storytelling with Data* by Cole Nussbaumer Knaflic draws the same line. **Exploratory** analysis is you searching the data for what is there. **Explanatory** analysis is you showing one finding to someone else.

	Exploring	Explaining
Like	Searching a messy cupboard for lost keys	Telling your family "the keys are on the hook by the door"
Who it is for	You	Someone else, who is short of time
How many charts	As many as it takes	As few as possible, ideally one per point
Done when	You know what is happening	They know what to do

Example. An analyst digs through 1,20,000 Rapido rides and finds that in Bengaluru, cancellations rose from 8% to 14% in two weeks, mostly between 6 and 9 p.m. That is exploring. The Ops head has five minutes before a meeting. She does not want 30 charts. She wants three answers: where is it slipping, how bad is it, and what should I do? Giving her those answers is explaining, and a dashboard is one of the best tools for it.

2. What a dashboard is, and what it is not

A dashboard is **one screen, for one person, to make one decision**. Think of a car's dashboard. It shows speed, fuel and warning lights. It does not show every detail of the engine. It shows what the driver needs *right now*.

A **report** explains a past period in detail, like a school report card at the end of the term. A dashboard is a quick check you can open any morning, like the speedometer. Both are useful, but they are different jobs.

Here is the same Rapido data on two different dashboards for the Ops head:

	Bad dashboard	Good dashboard
Who is it for?	"Everyone"	The Ops head
How many numbers?	25 charts, all the same size	4 key numbers, 1 trend, 1 city table
After looking, she...	Is not sure what to do	Knows which city to call first
How does it update?	Someone pastes new numbers every Monday	It reads fresh data by itself
Can she explore?	No, it is a fixed picture	Yes, she picks a city or a week from a list

The good dashboard shows *less*, and that is the point. Every extra chart makes the important one harder to find.

3. Plan before you build: five questions

Before you open Google Sheets, answer five questions on paper. It takes ten minutes and saves hours of rebuilding.

1. **Who is it for?** One person or role, such as "the Ops head", not "the team".
2. **What decision will they make?** Write it as a question: "Which city needs more captains this week?"
3. **Which 3 to 5 numbers help that decision?** Pick them from the metrics of that role. If a number would not change the decision, leave it out.
4. **What picture shows each number best?** A big number for a total, a line for change over time, bars to compare cities.
5. **How will they use it?** Which filters do they need (city, week)? How often does the data refresh? Who can only view, and who can edit?

Here is the recipe filled in for the Rapido Ops head:

Question	Rapido Ops head
Who is it for?	The Ops head, who runs rides across all cities
What decision?	Which city needs attention first this week?
Which numbers?	Rides completed, cancellation %, average fare, active captains
Which pictures?	Four big-number tiles, a weekly line for rides, bars for rides by city
How used?	Dropdowns for city and week, refreshes daily, view-only for everyone except the analyst

4. Your role picks the tiles

The same company needs different dashboards, because each role asks a different question. A **KPI tile** (also called a scorecard) is a big number on the dashboard, such as "Cancellation rate: 14%". Your role decides which tiles you put on top. The metrics below come from the metrics table in Pre-read 1.

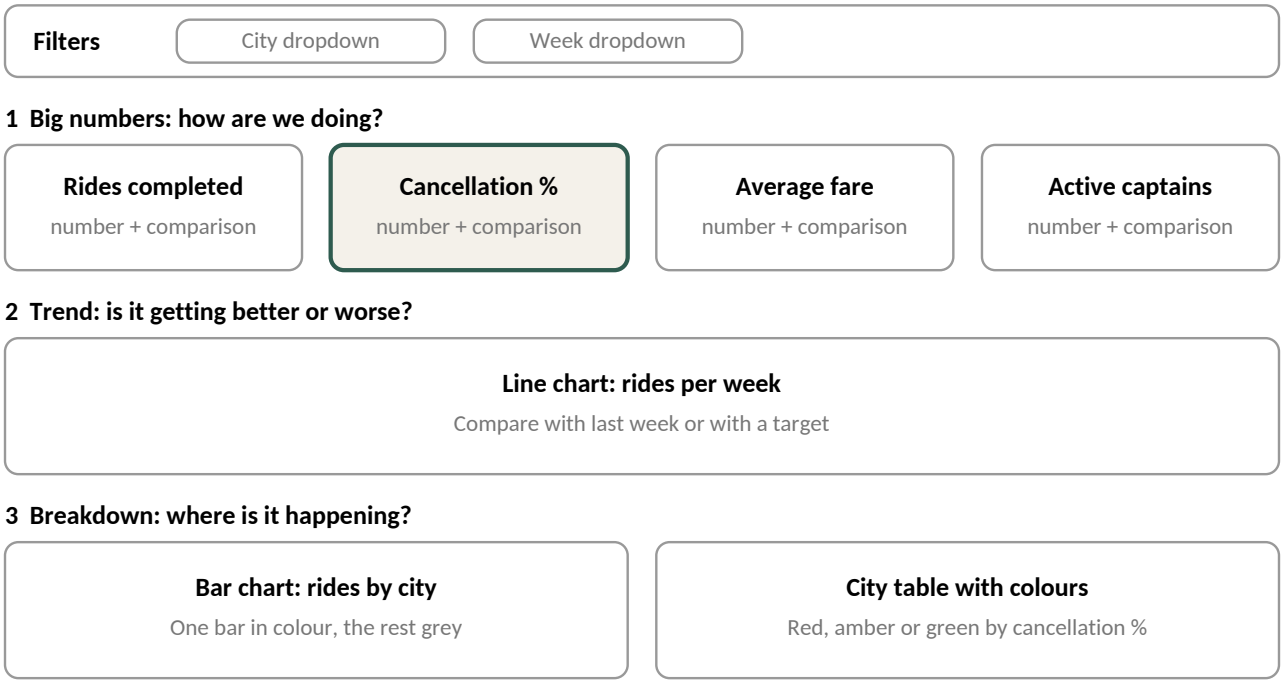
Role	Question the dashboard answers	KPI tiles	Main picture
Founder's Office	Is the business healthy?	Net revenue, contribution margin per order, burn rate, runway	Line: monthly net revenue
Chief of Staff	Are we on track with our goals?	Goals on track %, budget used vs planned, milestones on time	A red-amber-green list of goals
Program Manager	Is the project on time?	Milestones on time %, budget used vs planned, open risks	A timeline of milestones
Product Manager	Where do users drop off?	Conversion rate, activation rate, feature adoption	A funnel of steps
Growth Manager	Are customers coming back?	New customers, repeat rate, cohort retention (M1)	Lines: retention by signup month
Acquisition Manager	Which channel is worth the money?	CAC, ROAS, conversion rate	Bars: CAC by channel
Category Manager	Which products earn their shelf space?	Sales, gross margin %, availability	Bars: margin by product
Demand Planner	What should we order, and how much?	Days of cover, availability, forecast accuracy	Lines: forecast vs actual sales

Notice that no role needs all the metrics. Each picks three or four, and the picture that fits its question.

5. Anatomy of a dashboard

The screen: read top to bottom. Eyes land on the top-left first, so the most important numbers go there. Then comes the trend, then the detail.

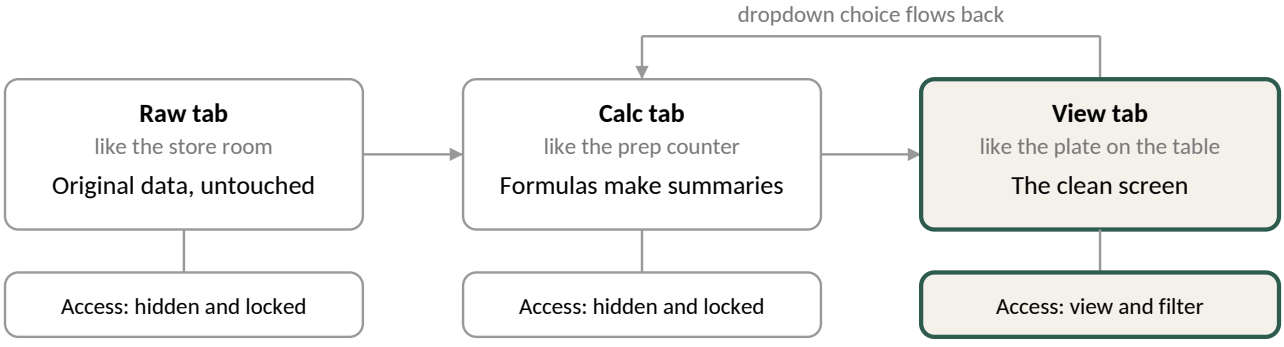
Read a dashboard top to bottom: numbers first, then trend, then detail



The numbers answer "how are we doing?", the trend answers "is it getting better or worse?", and the breakdown answers "where is it happening?". The one tile in colour is the one the reader should notice first.

Behind the screen: three tabs. Think of a restaurant kitchen. Ingredients wait in the store room, the chef chops and mixes them at the prep counter, and only the finished plate goes to the table. Guests never walk into the kitchen. A good Google Sheets dashboard works the same way.

Three tabs keep the dashboard clean and safe



The viewer's dropdown choice on the view tab flows back to the calc tab, which recalculates. Only the view tab is open to everyone.

This split has three payoffs. When new data arrives, you only replace the raw tab and nothing else breaks. Mistakes are easy to find, because all the formulas live in one place. And you can lock what must not be touched while still sharing the screen.

6. Reading numbers correctly

A dashboard is only as good as the way people read it. Reading data well is called **data interpretation**: working out what the numbers really mean, before you act. Here is one week of Rapido rides to practise on.

City	Rides last week	Rides this week	Change
Bengaluru	50,000	45,000	-10%
Chennai	30,000	31,500	+5%
Delhi	40,000	40,400	+1%
Hyderabad	25,000	24,500	-2%
All cities	1,45,000	1,41,400	-2.5%

Seven habits will keep you from reading it wrong.

1. **A number alone means nothing.** "Cancellations are 14%" says little. Compared with 8% last week, it is a problem. Always put a comparison next to it: last week, a target, or another city.
2. **Totals hide stories.** The last row says rides are down only 2.5%, which sounds fine. Break it by city and Bengaluru is down 10%. Always ask, "what is hiding inside this total?"
3. **Look at the percent and the real number together.** Chennai is +5%, which is 1,500 extra rides. Bengaluru is -10%, which is 5,000 lost rides. A city that goes from 20 rides to 30 is +50%, but it is only 10 rides.
4. **Averages can fool you.** Say 9 rides cost ₹50 and one ride costs ₹950. The average is ₹140, yet a typical ride is ₹50. Check the **median** (the middle value when you line them up in order) and look for an **outlier** (one value far away from all the rest).
5. **Small samples are shaky.** If 3 of 5 rides were cancelled, that is 60%, but five rides prove nothing. Ask, "how many rides is this based on?"
6. **Moving together is not the same as causing.** Cancellations rose the same week the app screen changed. Maybe the screen caused it, or maybe it rained and captains stayed home. This is the difference between **correlation** (two things move together) and **causation** (one makes the other happen). Check the other explanations before you blame anything.
7. **Check the axis.** If a bar chart starts at 90 instead of 0, a tiny gap looks huge. Bars should always start at zero.

End every reading with "so what?". A number that doesn't lead to an action is just decoration.

7. Showing it well: six lessons from *Storytelling with Data*

The book *Storytelling with Data* by Cole Nussbaumer Knaflic teaches how to turn numbers into something people understand and act on. Its chapters give six lessons, explained here in plain words (chapter list from a book summary video on YouTube). Read this as a checklist for every dashboard you build.

Lesson	In plain words	In the Rapido dashboard
1. Understand the context	Before drawing anything, know who the audience is, what they must do, and what you want them to remember	The Ops head must decide which city gets more captains this week
2. Choose an effective visual	Pick the picture that fits the question (see the guide below)	A line for rides over time, bars to compare cities
3. Clutter is your enemy	Remove anything that does not help the message: heavy borders, extra colours, 3D effects, long labels. Every item makes the reader work harder	Grey, thin gridlines, no 3D, only the labels that matter
4. Focus attention	Use colour, size and position to say "look here first". Make everything quiet except the one thing that matters	All city bars in grey, Bengaluru in orange
5. Think like a designer	Keep it neat: aligned, with space to breathe, and a title that states the finding, not just the topic. Neat work earns trust	Title: "Bengaluru rides fell 10% this week", not "Rides by city"
6. Tell a story	Give it a beginning (what was normal), a middle (what changed) and an end (what to do)	Normal week, then the Bengaluru drop, then the action

The Big Idea. The book suggests writing your whole message as one sentence that says what you think and what is at stake. For our Ops head: "Bengaluru lost 5,000 rides this week and cancellations jumped from 8% to 14%, mostly at 6 to 9 p.m., so we should add captains at those hours first." If you cannot write this sentence, you are not ready to build the dashboard.

Choosing the right picture

Your question	Best picture	Rapido example
How big is it?	A big-number tile	Rides today: 20,000
How do a few groups compare?	Bar chart	Rides by city
How does it change over time?	Line chart	Rides per week
How do parts make up a whole?	Stacked bar	Completed vs cancelled rides in each city
Are two numbers related?	Scatter chart (one dot per ride)	Distance vs fare
Which items need attention?	Colour-coded cells: red, amber, green	City table shaded by cancellation %

Three things to avoid: a pie chart with many slices (nobody can compare them), 3D charts (they bend the truth), and a line chart for groups with no order, like cities (a line suggests a trend that is not there).

8. Making it interactive and safe in Google Sheets

You don't need to memorise any clicks now. Just meet the tools by what each one is *for*, so that in class you know which one to reach for.

Make it interactive

Tool	In plain words	Why a dashboard needs it
Dropdown (data validation)	A cell where you pick from a list instead of typing	The Ops head picks "Chennai" and the whole screen changes
Slicer	A set of filter buttons that sits on top of a chart or table	Filter by vehicle type without touching any formulas
Named range	A nickname for a block of cells, like "city_list" instead of A2:A500	Formulas become easier to read and harder to break
Conditional formatting	Cells that change colour by themselves when a rule is met	Cancellation % turns red above 12%, so problems jump out
Sparkline	A tiny line chart that lives inside one cell	Show each city's weekly trend next to its name
Scorecard chart	A big number with a small comparison underneath	A KPI tile such as "Rides: 1,41,400, down 2.5%"

Make it safe

A dashboard is shared with people who did not build it. They should be able to look and filter, but not break the formulas.

Tool	In plain words	Why a dashboard needs it
Share settings	Decide who can **view** (look only), **comment** (look and leave notes) or **edit** (change anything)	Most people get view access; only the builder can edit
Protected sheet or range	Locks cells so that only chosen people can change them	The formulas stay safe even if someone clicks around
Hidden tabs	Tabs that are tucked away from the viewer	Hide the messy working tabs, show only the clean screen
IMPORTRANGE	A formula that pulls data from a different Google Sheet file	Keep the big raw data in one file and the dashboard in another

9. Jargon buster

New words from this pre-read, in plain language. Metric words such as CAC, ROAS and churn are in the Jargon buster of Pre-read 1.

Word	What it means	Tiny example
Dashboard	One screen that shows the few numbers one person needs for one decision	A car's speedometer and fuel gauge
KPI	Key Performance Indicator: a metric the team agreed matters most	Rides completed per day
KPI tile (scorecard)	A big number on a dashboard, often with a small comparison	"Cancellation rate: 14%"
Report	A detailed look at a past period	A school report card

Word	What it means	Tiny example
Stakeholder	Anyone who uses your analysis or is affected by it	The Ops head
Exploratory analysis	Digging through data to find what is there	Searching a cupboard for lost keys
Explanatory analysis	Showing one clear finding to others so they can act	"The keys are on the hook"
Data interpretation	Working out what the numbers really mean	Spotting that a total hides a falling city
Median	The middle value when numbers are lined up in order	For 50, 50, 50, 950 the middle is 50
Outlier	One value far away from all the others	A ₹950 ride among ₹50 rides
Sample	The group of items you looked at	5 rides out of 1,00,000
Correlation	Two things move together	Rain and cancellations both go up
Causation	One thing makes the other happen	Heavy rain makes captains stay home
Big Idea	Your whole message in one sentence: what you think and what is at stake	"Bengaluru lost 5,000 rides, so add captains at 6 to 9 p.m."
Clutter	Anything on a chart that does not help the message	Heavy borders, 3D effects
Red-amber-green (RAG)	Colours that show status: red is bad, amber is watch, green is good	A city shaded red for high cancellations
Raw tab	The tab holding the original data, untouched	The full list of rides
Calc tab	The tab where formulas turn raw data into summary numbers	Rides per city per week
View tab	The clean tab people actually look at	The dashboard screen
Dropdown	A cell where you pick from a list	Choosing a city
Slicer	Filter buttons attached to a chart or table	Buttons for Bike, Auto, Cab
Named range	A nickname for a block of cells	city_list for A2:A500
Conditional formatting	Cells that change colour by themselves when a rule is met	Red when cancellation % is above 12%
Sparkline	A tiny line chart inside one cell	A week of rides next to a city's name
Viewer / Commenter / Editor	Sharing levels: look only, look and comment, or change anything	Most people get Viewer
Protected range	Cells locked so only chosen people can change them	The formulas on the calc tab
IMPORTRANGE	A formula that pulls data from another Google Sheet file	Raw data in one file, dashboard in another